



Time: 5 minutes

Outcome: Explain the basic limitation on combining individual rankings into a social ranking.

Difficulty: Intermediate

Arrow's Impossibility Theorem

THE CORE IDEA

With at least three alternatives, Arrow's theorem shows that no ranked-choice social decision rule can guarantee all of a specified set of desirable conditions at once. The result identifies a tradeoff among aggregation properties; it does not prove that democracy or every voting system is impossible.

HOW TO RECOGNIZE IT

- The problem combines individual rankings into a complete and consistent social ranking.
- Desirable conditions include respecting unanimity and avoiding control by one dictator.
- Independence of irrelevant alternatives limits how an outside option may change a pairwise ranking.
- The theorem concerns simultaneous guarantees under its conditions, not whether voting can occur.

WATCH OUT

Do not conclude that no voting system works or that democracy is impossible. The theorem applies to ranked aggregation and the simultaneous satisfaction of specified conditions.

WORKED EXAMPLE: DESIRABLE RULES IN TENSION

Desirable guarantees and their meanings

	Meaning
Unrestricted rankings	Allow any consistent individual rankings.
Consistent social ranking	Rank every pair without a group cycle.
Unanimity	Respect a comparison everyone agrees on.
No dictator	No one voter always determines the ranking.
Independence	If each voter keeps A versus B unchanged, the social A-versus-B ranking stays unchanged.

A community wants all five guarantees in the table. With at least three alternatives, Arrow's theorem says no rule combining unrestricted individual rankings can always provide every guarantee. The introductory lesson is a design tradeoff among reasonable goals, not that elections cannot produce winners.

CHECK YOURSELF

Which is accurate: Arrow identifies incompatible guarantees under stated conditions, or proves that no election can produce a winner?